

Malicious URL/Phishing Detector

Algorithm Implementation

We made a Bloom Filter class, which has the methods `add()`, `check()`, `_hashes()`, `get_fill_ratio()`, `get_num_bits_set()`, `save_bit_array()`, and `load_bit_array()`. Since we need to run the Bloom filter twice, once offline to set up the bit array and once online to test functionality, this makes sure the code is reusable and clean.

We implemented the MurmurHash3 function as our hash function, as we had many malicious URLs in the database and therefore needed an effective hash function to maximize space and efficiency. We then called the hash function within the Bloom filter class function `_hashes()`, returning each of the indices.

`add()` is a function to add the given URL into the bit array, which will only be called during the setup when given a list of entirely malicious URLs. `check()` is only called during the online streaming test part of the code, which will use the hash functions to check whether or not a URL is in the array and return a boolean.

`save_bit_array()` and `load_bit_array()` are necessary functions because we need to maintain a global bit array for both online and offline. These functions allow us to save the bit array created offline to a file, and then load that file when we are testing online to prevent delays in checking a URL.

Evaluation

Out of 361,933 tested URLs, all 8,062 malicious URLs were correctly identified as malicious.

Only 928 URLs were falsely identified as malicious, meaning there was a false positive rate of 0.0026%. There were also 0 false negatives, meaning the algorithm works as expected.